

# C1 Geometry — Lesson Plan

IGCSE Mathematics 0580 | Concept Guide

## C1\_Q1: Angles: Types and Basic Properties

### Hook

Have you ever noticed how a simple door hinge creates different angles as you open the door? From tightly closed ( $0^\circ$ ) to fully open ( $180^\circ$  or more), every position represents a different type of angle. Understanding angles is the foundation of all geometry — and in your IGCSE exam, angle problems appear in almost every paper!

### Prerequisites

- Basic understanding of points, lines, and rays
- Ability to measure with a ruler
- Understanding of the number line and degrees as a unit of measurement

### Core Explanation

An **angle** is formed when two rays (or line segments) meet at a common endpoint called the **vertex**. We measure angles in **degrees** ( $^\circ$ ).

#### Types of Angles

| Angle Type    | Measure                          | Description                                |
|---------------|----------------------------------|--------------------------------------------|
| Acute         | $0^\circ < \theta < 90^\circ$    | A small, sharp angle                       |
| Right angle   | $\theta = 90^\circ$              | A quarter-turn, marked with a small square |
| Obtuse        | $90^\circ < \theta < 180^\circ$  | A wide, blunt angle                        |
| Straight line | $\theta = 180^\circ$             | A half-turn, forms a straight line         |
| Reflex        | $180^\circ < \theta < 360^\circ$ | A large angle, more than a straight line   |
| Full rotation | $\theta = 360^\circ$             | A complete turn                            |

#### Angle Notation

We name angles using three points: the vertex is always the middle letter. For example,  $\angle ABC$  means the angle at vertex B, formed by rays BA and BC. We can also use a single letter if the vertex is clear (e.g.,  $\angle A$ ) or a number (e.g.,  $\angle 1$ ).

#### Measuring Angles with a Protractor

- A protractor measures angles from  $0^\circ$  to  $180^\circ$ . To measure an angle:
1. Place the centre hole of the protractor on the vertex.
  2. Align the  $0^\circ$  line with one ray of the angle.
  3. Read the measurement where the other ray crosses the protractor scale (use the inner scale for angles

opening to the right, outer for angles opening to the left).

### ### Basic Angle Properties

Three fundamental angle facts you must memorise:

**1. Angles at a point sum to  $360^\circ$**

When several angles meet at a single point, their measures add up to  $360^\circ$ .

$$a + b + c + d = 360^\circ$$

**2. Angles on a straight line sum to  $180^\circ$**

If two or more angles lie on a straight line, they add up to  $180^\circ$ .

$$p + q + r = 180^\circ$$

**3. Vertically opposite angles are equal**

When two straight lines intersect, the angles directly across from each other are equal.

$$\angle A = \angle C \quad \text{and} \quad \angle B = \angle D$$

## Worked Example

**Problem:** In the diagram below, three angles meet at a point on a straight line.  $\angle AOB = 47^\circ$  and  $\angle BOC = 83^\circ$ . Find  $\angle COD$ .

(Imagine a straight line A-O-D with ray OB and OC meeting at O between them.)

**Step 1:** Identify what we know.

Points A, O, D lie on a straight line, so angles on a straight line sum to  $180^\circ$ .

$$\angle AOB + \angle BOC + \angle COD = 180^\circ$$

**Step 2:** Substitute the known values.

$$47^\circ + 83^\circ + \angle COD = 180^\circ$$

**Step 3:** Simplify.

$$130^\circ + \angle COD = 180^\circ$$

**Step 4:** Solve for  $\angle COD$ .

$$\angle COD = 180^\circ - 130^\circ = 50^\circ$$

**Answer:**  $\angle COD = 50^\circ$

## Common Mistakes

**X Using the wrong scale on a protractor (reading the outer scale when the inner scale should be used)**

OK Fix: Always check whether the angle is acute ( $< 90^\circ$ ) or obtuse ( $> 90^\circ$ ) first. If you read  $120^\circ$  for an acute-l

**X Confusing vertically opposite angles with adjacent angles on a straight line.**

OK Fix: Vertically opposite angles are across from each other at an intersection (like opposite corners of an X).

**X Forgetting that 'angles at a point' means all angles around a single point sum to  $360^\circ$ , not  $180^\circ$ .**

OK Fix: Remember: a full circle is  $360^\circ$ . If the angles fill the space around a point, they sum to  $360^\circ$ . Only if the

**X Labelling angles incorrectly — writing  $\angle ABC$  when the vertex should be the middle letter but put**

OK Fix: The vertex is always the MIDDLE letter.  $\angle ABC$  has vertex B. Check: the middle letter of the three is the

## Summary

- Angles are measured in degrees ( $^\circ$ ) and classified as acute ( $0-90^\circ$ ), right ( $90^\circ$ ), obtuse ( $90-180^\circ$ ), straight ( $180^\circ$ ), or reflex ( $180-360^\circ$ ).
- Angle notation uses three points with the vertex as the middle letter, e.g.,  $\angle ABC$  means the angle at point B.
- A protractor is used to measure angles; check whether to use the inner or outer scale.
- Angles at a point always sum to  $360^\circ$ .
- Angles on a straight line always sum to  $180^\circ$ .
- Vertically opposite angles formed by intersecting lines are always equal.

## C1\_Q2: Parallel Lines and Transversals

### Hook

When you look at a railway track stretching into the distance, the two steel rails never meet — they are a real-world example of parallel lines. Now imagine a train crossing those tracks at a diagonal: the railway crossing creates a pattern of angles that mathematicians have studied for centuries. Once you decode this pattern, you can find missing angles without measuring anything!

### Prerequisites

- C1\_Q1: Types of angles and basic angle properties
- Angles on a straight line sum to  $180^\circ$
- Vertically opposite angles are equal
- Understanding of the terms 'alternate', 'corresponding', and 'interior'

### Core Explanation

**Parallel lines** are lines in the same plane that never meet. We denote them with the symbol  $\parallel$ . A **transversal** is a line that crosses two or more parallel lines.

When a transversal cuts two parallel lines, it creates eight angles at the two intersection points. These angles have special relationships:

#### 1. Corresponding Angles (F-pattern)

Corresponding angles are in the same relative position at each intersection. They form an 'F' shape (possibly rotated or reflected).

- They are **equal**.

If  $AB \parallel CD$ , then  $\angle 1 = \angle 5$ ,  $\angle 2 = \angle 6$ ,  $\angle 3 = \angle 7$ ,  $\angle 4 = \angle 8$

### 2. Alternate Angles (Z-pattern)

Alternate angles are inside the parallel lines but on opposite sides of the transversal. They form a 'Z' shape (or a backwards 'Z').

- They are **equal**.

If  $AB \parallel CD$ , then  $\angle 3 = \angle 6$  and  $\angle 4 = \angle 5$

### 3. Co-interior (Allied) Angles (C-pattern)

Co-interior angles are inside the parallel lines and on the same side of the transversal. They form a 'C' shape.

- They sum to  $180^\circ$  (they are **supplementary**).

If  $AB \parallel CD$ , then  $\angle 3 + \angle 5 = 180^\circ$  and  $\angle 4 + \angle 6 = 180^\circ$

### Summary Table

| Relationship         | Shape | Property           |
|----------------------|-------|--------------------|
| Corresponding        | F     | Equal              |
| Alternate            | Z     | Equal              |
| Co-interior (Allied) | C     | Sum to $180^\circ$ |

### Identifying Parallel Lines

You can also work backwards: if you know certain angle relationships exist, you can prove the lines are parallel.

- If a pair of corresponding angles are equal, the lines are parallel.
- If a pair of alternate angles are equal, the lines are parallel.
- If a pair of co-interior angles sum to  $180^\circ$ , the lines are parallel.

## Worked Example

**Problem:** In the diagram,  $AB \parallel CD$ . The transversal EF cuts both lines.  $\angle AEF = 63^\circ$ . Find  $\angle EFD$  and  $\angle FEB$ .

(Imagine line AB above line CD, with transversal EF crossing from left to right, slanting downwards.)

**Step 1:** Identify the relationship between  $\angle AEF$  and  $\angle EFD$ .

- $\angle AEF$  is above line AB on the left of the transversal.
- $\angle EFD$  is below line CD on the right of the transversal.
- These are **alternate angles** (Z-pattern).
- Therefore they are equal.

**Step 2:** Calculate  $\angle EFD$ .

$$\angle EFD = \angle AEF = 63^\circ$$

**Step 3:** Find  $\angle FEB$ .

-  $\angle AEF$  and  $\angle FEB$  lie on a straight line (AB).

- Angles on a straight line sum to  $180^\circ$ .

$$\angle AEF + \angle FEB = 180^\circ$$

$$63^\circ + \angle FEB = 180^\circ$$

$$\angle FEB = 180^\circ - 63^\circ = 117^\circ$$

**Check:**  $\angle FEB$  and the co-interior angle at the lower intersection should sum to  $180^\circ$ .  
Indeed,  $\angle FEB = 117^\circ$  corresponds to an angle that is  $180^\circ - 63^\circ = 117^\circ$  at the lower line.

**Answer:**  $\angle EFD = 63^\circ$  and  $\angle FEB = 117^\circ$

## Common Mistakes

**X Confusing alternate and corresponding angles — assuming all angles formed by a transversal are equal.**

OK Fix: Trace the F and Z shapes carefully. Corresponding angles are in the same position at each intersection.

**X Forgetting that co-interior (allied) angles sum to  $180^\circ$  — not that they are equal.**

OK Fix: Remember the 'C' shape for co-interior angles. C has curved lines that suggest they 'co-operate' to make a straight line.

**X Applying parallel line rules to lines that are not marked as parallel.**

OK Fix: Do not assume lines are parallel unless they are marked with arrowheads ( $\parallel$ ) or stated in the question.

**X Thinking all acute angles formed by a transversal across parallel lines are equal (ignoring the obtuse angles).**

OK Fix: When a transversal cuts parallel lines, the acute angles are all equal, the obtuse angles are all equal, and corresponding angles are equal.

## Summary

- Parallel lines ( $\parallel$ ) never meet; a transversal is a line that crosses two or more parallel lines.
- Corresponding angles (F-pattern) are equal when lines are parallel.
- Alternate angles (Z-pattern) are equal when lines are parallel.
- Co-interior / allied angles (C-pattern) sum to  $180^\circ$  when lines are parallel.
- You can prove lines are parallel by checking if any of these relationships holds between the angles.
- Only apply these rules when lines are explicitly marked as parallel or proven to be parallel.

## C1\_Q3: Polygons: Interior and Exterior Angles

### Hook

Have you ever wondered why honeybees build their hives using hexagons? Or why soccer balls are made of pentagons and hexagons? The answer lies in the mathematics of polygons! Polygons are everywhere — from stop signs (octagons) to the Pyramids of Giza (triangles). Understanding their angle properties

helps architects, engineers, and even nature decide the best shapes to use.

## Prerequisites

- C1\_Q1: Types of angles and basic angle properties
- C1\_Q2: Parallel lines and transversals
- Understanding of triangles and their angle sum
- Basic algebra skills for solving equations

## Core Explanation

A **polygon** is a closed 2D shape made of straight line segments. Polygons are named by their number of sides.

| Number of Sides | Name          |
|-----------------|---------------|
| 3               | Triangle      |
| 4               | Quadrilateral |
| 5               | Pentagon      |
| 6               | Hexagon       |
| 7               | Heptagon      |
| 8               | Octagon       |
| 9               | Nonagon       |
| 10              | Decagon       |

### Regular vs Irregular Polygons

- **Regular polygon**: All sides are equal in length AND all interior angles are equal.

- **Irregular polygon**: Sides and/or angles are not all equal.

### Interior Angle Sum

The interior angles of any polygon sum to a value that depends only on the number of sides.

$$\text{Sum of interior angles} = (n - 2) \times 180^\circ$$

where  $n$  is the number of sides.

**Why?** A polygon can be divided into  $(n-2)$  triangles by drawing diagonals from one vertex. Each triangle has an angle sum of  $180^\circ$ , so the total sum is  $(n-2) \times 180^\circ$ .

### Exterior Angle Sum

For ANY polygon, regardless of whether it is regular or irregular:

$$\text{Sum of exterior angles} = 360^\circ$$

An exterior angle is formed by extending one side of the polygon. At each vertex, the interior angle + exterior angle =  $180^\circ$  (they lie on a straight line).

### ### Regular Polygon Formulas

For a regular polygon with  $n$  sides:

$$\text{Each interior angle} = \frac{(n-2) \times 180^\circ}{n}$$

$$\text{Each exterior angle} = \frac{360^\circ}{n}$$

$$\text{Interior angle} + \text{Exterior angle} = 180^\circ$$

### ### Finding the Number of Sides Given an Angle

If you know one interior or exterior angle of a regular polygon, you can find  $n$ .

$$\text{Using exterior angle: } n = \frac{360^\circ}{\text{exterior angle}}$$

$$\text{Using interior angle: } \text{First find exterior angle} = 180^\circ - \text{interior angle}, \text{ then } n = \frac{360^\circ}{\text{exterior angle}}$$

## Worked Example

**Problem 1:** Find the sum of interior angles of a regular pentagon, and the size of each interior angle.

**Step 1:** Identify  $n = 5$  (pentagon has 5 sides).

**Step 2:** Calculate the sum of interior angles.

$$\text{Sum} = (n - 2) \times 180^\circ = (5 - 2) \times 180^\circ = 3 \times 180^\circ = 540^\circ$$

**Step 3:** For a regular pentagon, divide by  $n$  to find each interior angle.

$$\text{Each interior angle} = \frac{540^\circ}{5} = 108^\circ$$

**Answer:** Sum =  $540^\circ$ , each interior angle =  $108^\circ$ .

**Check using exterior angle:** Exterior angle =  $360^\circ / 5 = 72^\circ$ . Interior =  $180^\circ - 72^\circ = 108^\circ$ . ✓

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**Problem 2:** A regular polygon has interior angles of  $150^\circ$ . How many sides does it have?

**Step 1:** Find the exterior angle.

$$\text{Exterior angle} = 180^\circ - 150^\circ = 30^\circ$$

**Step 2:** Use the exterior angle formula to find  $n$ .

$$n = \frac{360^\circ}{30^\circ} = 12$$

**\*\*Answer:\*\*** The polygon has 12 sides (a regular dodecagon).

## Common Mistakes

**X Using the formula  $(n-2) \times 180^\circ$  to find each interior angle of a regular polygon, for**

OK Fix: The formula  $(n-2) \times 180^\circ$  gives the TOTAL sum of interior angles. For a regular polygon, you

**X Thinking the sum of exterior angles is  $(n-2) \times 180^\circ$  (like interior angles) or some other**

OK Fix: The sum of exterior angles is ALWAYS  $360^\circ$  for any convex polygon, no matter how many sides

**X Confusing interior and exterior angles — solving for the wrong one when given a problem about**

OK Fix: Draw a small diagram showing the extended side. The exterior angle is OUTSIDE the shape. Remember

**X Applying regular polygon formulas to irregular polygons — assuming all angles are equal when t**

OK Fix: You can only use  $\frac{(n-2) \times 180}{n}$  for each interior angle if the polygon is REGULAR. For i

## Summary

- A polygon is a closed shape with straight sides; regular polygons have equal sides and equal angles.
- Sum of interior angles =  $(n-2) \times 180^\circ$ , where  $n$  is the number of sides.
- Sum of exterior angles =  $360^\circ$  for any convex polygon.
- For a regular polygon: each interior angle =  $\frac{(n-2) \times 180^\circ}{n}$ , each exterior angle =  $\frac{360^\circ}{n}$ .
- Interior and exterior angles at the same vertex are supplementary (sum to  $180^\circ$ ).
- Given one interior or exterior angle of a regular polygon, you can find the number of sides using  $n = \frac{360^\circ}{\text{exterior angle}}$ .

## C1\_Q4: Triangles: Types, Angle Properties, and Congruence

### Hook

The triangle is the simplest of all polygons — just three sides — yet it is the strongest shape in engineering and construction. From the Eiffel Tower to bridge trusses, triangles provide stability because they cannot be deformed without changing the length of a side. In this topic, you will discover the secrets hidden inside every triangle: how its angles behave, how it can be classified, and how we prove two triangles are identical copies of each other.

### Prerequisites

- C1\_Q1: Types of angles and basic angle properties
- C1\_Q2: Parallel lines (for exterior angle proof)
- C1\_Q3: Basic polygon concepts
- Angle notation:  $\angle ABC$



## Core Explanation

### Types of Triangles by Sides

| Triangle Type | Side Properties | Notation |

| ---|---|---|

| **Scalene** | All three sides are different lengths | No equal sides |

| **Isosceles** | Two sides are equal | The equal sides are called 'legs'; the third is the 'base'. Base angles are equal. |

| **Equilateral** | All three sides are equal | All angles are  $60^\circ$  |

### Types of Triangles by Angles

| Triangle Type | Angle Properties |

| ---|---|

| **Acute-angled** | All three angles are less than  $90^\circ$  |

| **Right-angled** | One angle equals exactly  $90^\circ$  (marked with a square) |

| **Obtuse-angled** | One angle is greater than  $90^\circ$  |

### Key Angle Theorems

**1. Angle sum of a triangle =  $180^\circ$**

The three interior angles of any triangle sum to  $180^\circ$ .

$$\angle A + \angle B + \angle C = 180^\circ$$

**2. Exterior angle theorem**

An exterior angle of a triangle equals the sum of the two opposite interior angles.

$$\text{Exterior angle} = \text{Sum of opposite interior angles}$$

$$\angle ACD = \angle CAB + \angle ABC$$

(Imagine triangle ABC with side BC extended to D. The exterior angle at C is  $\angle ACD$ .)

### Congruence

Two triangles are **congruent** if they are exactly the same shape and size — all corresponding sides equal and all corresponding angles equal. We write  $\triangle ABC \cong \triangle DEF$ .

We do NOT need to check all six measurements. Four tests tell us enough:

| Test | Condition | Meaning |

| ---|---|---|

| **SSS** | Three sides equal | Side-Side-Side |

| **SAS** | Two sides and the **included** angle equal | Side-Angle-Side |

| **ASA** | Two angles and the **included** side equal | Angle-Side-Angle |

| **RHS** | Right angle, Hypotenuse, and one other side equal | Right-Hypotenuse-Side (only for right triangles) |

**Important:** There is NO 'ASS' test. Two sides and a non-included angle does NOT guarantee congruence (it can produce two different triangles — the 'ambiguous case').

## Worked Example

**Problem:** In  $\triangle ABC$ ,  $\angle A = 53^\circ$  and  $\angle B = 71^\circ$ . Find  $\angle C$ . Also, find the exterior angle at C.

**Step 1:** Use the angle sum of a triangle.

$$\angle A + \angle B + \angle C = 180^\circ$$

$$53^\circ + 71^\circ + \angle C = 180^\circ$$

$$124^\circ + \angle C = 180^\circ$$

$$\angle C = 56^\circ$$

**Step 2:** Find the exterior angle at C.

The exterior angle is the angle outside the triangle formed by extending side BC past C. By the exterior angle theorem:

$$\text{Exterior angle at C} = \angle A + \angle B = 53^\circ + 71^\circ = 124^\circ$$

**Check:** Interior angle C ( $56^\circ$ ) + Exterior angle ( $124^\circ$ ) =  $180^\circ$ . They form a straight line. ✓

**Answer:**  $\angle C = 56^\circ$ , exterior angle at C =  $124^\circ$ .

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**Problem (Congruence):**  $\triangle PQR$  and  $\triangle XYZ$  have  $PQ = XY = 5$  cm,  $PR = XZ = 7$  cm, and  $\angle QPR = \angle YXZ = 40^\circ$ . Are the triangles congruent? If so, by which test?

**Step 1:** Identify what we know.

- Two sides:  $PQ = XY$  and  $PR = XZ$
- The angle between those two sides:  $\angle QPR = \angle YXZ$
- The angle  $\angle QPR$  is between sides  $PQ$  and  $PR$ .
- The angle  $\angle YXZ$  is between sides  $XY$  and  $XZ$ .

**Step 2:** Check the congruence test.

- Two sides and the included angle: this matches the **SAS** test.

**Answer:** Yes,  $\triangle PQR \cong \triangle XYZ$  by SAS.

## Common Mistakes

**X Assuming that 'two sides and any angle' is sufficient for congruence (the ASS/SSA misconception)**

OK Fix: The angle **MUST** be the **INCLUDED** angle — the angle between the two known sides. SAS works; SSA does not.

**X Thinking the exterior angle equals just one opposite interior angle, or all three angles combined**

OK Fix: The exterior angle equals the SUM of the TWO opposite interior angles (the two angles that are NOT adjacent).

**X Forgetting the isosceles triangle property — if two sides are equal, the base angles are equal, but not vice versa**

OK Fix: In an isosceles triangle, the two equal sides meet at the apex. The base angles are at the ends of the third side.

**X Confusing congruence with similarity — stating triangles are congruent when they are only the same shape**

OK Fix: Congruent means same shape AND same size (all side lengths equal). Similar means same shape but not necessarily same size.

## Summary

- Triangles are classified by sides (scalene, isosceles, equilateral) and by angles (acute, right, obtuse).
- The sum of interior angles in any triangle is  $180^\circ$ .
- An exterior angle of a triangle equals the sum of the two opposite interior angles.
- Congruence means identical shape and size — proven by SSS, SAS, ASA, or RHS.
- There is no ASS/SSA test for congruence; the angle must be the included angle.
- In an isosceles triangle, base angles opposite the equal sides are equal.

## C1\_Q5: Quadrilaterals: Properties and Classification

### Hook

Look around your room right now. Chances are, you see rectangles everywhere — windows, doors, screens, books. But did you know that a square is actually a special kind of rectangle, a rectangle is a special kind of parallelogram, and a parallelogram is a special kind of quadrilateral? These shapes form a family tree, each one inheriting properties from its parent. Understanding this hierarchy is the key to mastering quadrilaterals in your IGCSE exams!

### Prerequisites

- C1\_Q1: Angles and basic angle properties
- C1\_Q3: Basic polygon concepts
- C1\_Q4: Triangle properties and angle sum
- Parallel lines and angle relationships

### Core Explanation

A **quadrilateral** is any polygon with 4 sides. The sum of its interior angles is always  $360^\circ$  (since  $(4-2) \times 180^\circ = 360^\circ$ ).

#### Types of Quadrilaterals and Their Properties

##### 1. Square

- All four sides equal
- All four angles =  $90^\circ$

- Diagonals are equal in length
- Diagonals bisect each other at  $90^\circ$
- Diagonals bisect the angles
- 4 lines of symmetry
- Rotational symmetry of order 4

#### #### 2. Rectangle

- Opposite sides are equal and parallel
- All four angles =  $90^\circ$
- Diagonals are equal in length
- Diagonals bisect each other (but NOT at  $90^\circ$  unless it's a square)
- 2 lines of symmetry
- Rotational symmetry of order 2

#### #### 3. Rhombus

- All four sides equal
- Opposite sides are parallel
- Opposite angles are equal
- Diagonals bisect each other at  $90^\circ$
- Diagonals bisect the interior angles
- 2 lines of symmetry
- Rotational symmetry of order 2

#### #### 4. Parallelogram

- Opposite sides are equal and parallel
- Opposite angles are equal
- Adjacent angles sum to  $180^\circ$  (co-interior with parallel lines)
- Diagonals bisect each other
- Diagonals are NOT necessarily equal
- No lines of symmetry (in general)
- Rotational symmetry of order 2

#### #### 5. Trapezium (US: Trapezoid)

- One pair of opposite sides is parallel
- The non-parallel sides are called 'legs'
- If the legs are equal, it is an **isosceles trapezium** — then base angles are equal and diagonals are equal
- No lines of symmetry (unless isosceles — then 1 line)
- Rotational symmetry of order 1 (none)

#### #### 6. Kite

- Two pairs of adjacent equal sides
- One pair of opposite angles are equal (the angles between unequal sides)
- Diagonals intersect at  $90^\circ$
- One diagonal bisects the other
- 1 line of symmetry

- Rotational symmetry of order 1 (none)

### Quadrilateral Hierarchy

$\text{Quadrilateral} \supset \text{Trapezium} \supset \text{Parallelogram} \supset \begin{cases} \text{Rectangle} \\ \text{Square} \end{cases}$   
 $\text{Rhombus} \supset \text{Square}$

A square is a special rectangle, a special rhombus, and a special parallelogram. A rectangle is a special parallelogram. A rhombus is a special parallelogram.

### Diagonals Properties Summary

| Shape         | Diagonals Equal? | Diagonals Bisect?     | Diagonals Perpendicular? | Diagonals Bisect Angles? |
|---------------|------------------|-----------------------|--------------------------|--------------------------|
| Square        | ✓                | ✓                     | ✓                        | ✓                        |
| Rectangle     | ✓                | ✓                     | ✗                        | ✗                        |
| Rhombus       | ✗                | ✓                     | ✓                        | ✓                        |
| Parallelogram | ✗                | ✓                     | ✗                        | ✗                        |
| Trapezium     | ✗ (usually)      | ✗                     | ✗                        | ✗                        |
| Kite          | ✗                | One bisects the other | ✓                        | ✗                        |

## Worked Example

**Problem:** In parallelogram ABCD,  $\angle A = 73^\circ$ . Find all other angles. Also, if the diagonals intersect at O and  $AO = 4$  cm, find OC.

**Step 1:** Use properties of a parallelogram.

- Opposite angles are equal:  $\angle C = \angle A = 73^\circ$
- Adjacent angles are supplementary (co-interior with parallel lines):  $\angle A + \angle B = 180^\circ$

**Step 2:** Find  $\angle B$ .

$$73^\circ + \angle B = 180^\circ$$

$$\angle B = 107^\circ$$

**Step 3:** Use opposite angles equal.

$$\angle D = \angle B = 107^\circ$$

**Step 4:** Check:  $73^\circ + 107^\circ + 73^\circ + 107^\circ = 360^\circ$ . ✓

**Step 5:** Find OC.

- In a parallelogram, diagonals bisect each other.
- O is the midpoint of both diagonals.
- $AO = OC = 4$  cm.

**Answer:**  $\angle B = 107^\circ$ ,  $\angle C = 73^\circ$ ,  $\angle D = 107^\circ$ ,  $OC = 4$  cm.

## Common Mistakes

**X Thinking that all quadrilaterals with parallel sides are parallelograms — a trapezium has parallel sides**

OK Fix: Parallelogram: BOTH pairs of opposite sides parallel. Trapezium: exactly ONE pair of parallel sides. Check!

**X Assuming the diagonals of a rectangle are perpendicular (they are equal but not perpendicular, they are equal)**

OK Fix: Rectangle diagonals are EQUAL in length and BISECT each other but are NOT perpendicular. Perpendicular diagonals are a property of a rhombus.

**X Confusing the properties of a rhombus and a parallelogram — thinking a rhombus has unequal sides**

OK Fix: A rhombus IS a special parallelogram. It has ALL parallelogram properties PLUS: all sides equal, diagonals perpendicular.

**X Forgetting that a square inherits ALL properties of rectangles AND rhombuses — it's easy to miss some**

OK Fix: A square is at the bottom of the hierarchy. It has: equal sides (rhombus), right angles (rectangle), equal diagonals (rectangle), perpendicular diagonals (rhombus).

## Summary

- All quadrilaterals have interior angles summing to  $360^\circ$ .
- A square has all properties of rectangles and rhombuses: equal sides, right angles, equal and perpendicular diagonals that bisect angles.
- A rectangle has equal diagonals; a rhombus has perpendicular diagonals.
- A parallelogram has opposite sides parallel and equal, opposite angles equal, adjacent angles supplementary, and diagonals that bisect each other.
- A trapezium has exactly one pair of parallel sides; an isosceles trapezium has equal base angles and equal diagonals.
- A kite has two pairs of adjacent equal sides, perpendicular diagonals, and one axis of symmetry.

## C1\_Q6: Circle Theorems 1: Angles in Circles and Cyclic Quadrilaterals

### Hook

The circle is one of the most elegant shapes in mathematics — perfectly symmetrical and deceptively simple. But hidden within its circumference are some of the most beautiful geometric relationships ever discovered. Ancient Greek mathematicians like Thales and Euclid uncovered these theorems over 2000 years ago, and they still appear on every IGCSE paper today. The most surprising? The angle at the centre of a circle is always exactly twice the angle at the circumference — no matter where you draw them!

### Prerequisites

- C1\_Q1: Types of angles and basic angle properties
- C1\_Q2: Parallel lines (angles on a straight line)
- C1\_Q4: Triangle properties, isosceles triangles
- Basic understanding of circles: centre, radius, chord, arc, sector, segment

### Core Explanation

A circle has several key parts: the **centre** (O), **radius** (distance from centre to edge), **diameter** (a chord through the centre, twice the radius), **chord** (a line segment joining two points on the circumference).

circumference), **arc** (a curved part of the circumference), and **segment** (region bounded by a chord and an arc).

### Theorem 1: Angle in a Semicircle

An angle inscribed in a semicircle is always a **right angle** ( $90^\circ$ ).

If AB is a diameter of a circle and C is any point on the circumference (other than A and B), then:

$$\angle ACB = 90^\circ$$

### Theorem 2: Angle at the Centre is Twice the Angle at the Circumference

The angle subtended by an arc at the centre of the circle is **twice** the angle subtended by the same arc at any point on the circumference.

If A and B are points on the circumference and O is the centre, and C is any other point on the circumference on the same side of AB:

$$\angle AOB = 2 \times \angle ACB$$

This is the most powerful circle theorem — it connects any angle at the circumference to its corresponding angle at the centre.

### Theorem 3: Angles in the Same Segment are Equal

All angles subtended by the same chord (or arc) in the same segment of a circle are equal.

If points C and D both lie on the same arc (the same side of chord AB):

$$\angle ACB = \angle ADB$$

### Theorem 4: Cyclic Quadrilateral

A **cyclic quadrilateral** is a quadrilateral whose four vertices all lie on the circumference of a circle.

**Opposite angles in a cyclic quadrilateral sum to  $180^\circ$ .**

If ABCD is a cyclic quadrilateral:

$$\angle A + \angle C = 180^\circ$$

$$\angle B + \angle D = 180^\circ$$

This follows from Theorem 2: the two opposite angles subtend arcs that together make the whole circle ( $360^\circ$  at the centre, so half that at the circumference).

### Connection Between Theorems

Notice that Theorem 1 is actually a special case of Theorem 2! When the chord AB is a diameter,  $\angle AOB = 180^\circ$  (straight line through the centre). Then:

$$\angle ACB = \frac{1}{2} \times \angle AOB = \frac{1}{2} \times 180^\circ = 90^\circ$$

## Worked Example

**Problem:** In the diagram, O is the centre of the circle. A, B, and C are points on the circumference.  $\angle AOB = 110^\circ$  and  $\angle OBC = 25^\circ$ . Find  $\angle ACB$  and  $\angle OCB$ .

**Step 1:** Find  $\angle ACB$  using Theorem 2 (angle at centre is twice angle at circumference).

- $\angle AOB$  is the angle at the centre subtended by arc AB.
- $\angle ACB$  is the angle at the circumference subtended by the same arc AB.

$$\angle AOB = 2 \times \angle ACB$$

$$110^\circ = 2 \times \angle ACB$$

$$\angle ACB = 55^\circ$$

**Step 2:** Find  $\angle OCB$ .

- Triangle OBC is isosceles because  $OB = OC$  (both are radii).
- In an isosceles triangle, base angles are equal.
- $\angle OBC = \angle OCB$  (base angles)

$$\angle OCB = \angle OBC = 25^\circ$$

**Answer:**  $\angle ACB = 55^\circ$ ,  $\angle OCB = 25^\circ$ .

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**Problem:** ABCD is a cyclic quadrilateral.  $\angle A = 78^\circ$  and  $\angle B = 115^\circ$ . Find  $\angle C$  and  $\angle D$ .

**Step 1:** Use the cyclic quadrilateral theorem.

- Opposite angles sum to  $180^\circ$ .

$$\angle A + \angle C = 180^\circ$$

$$78^\circ + \angle C = 180^\circ$$

$$\angle C = 102^\circ$$

**Step 2:** Find  $\angle D$ .

$$\angle B + \angle D = 180^\circ$$

$$115^\circ + \angle D = 180^\circ$$

$$\angle D = 65^\circ$$

**Check:**  $\angle A + \angle B + \angle C + \angle D = 78^\circ + 115^\circ + 102^\circ + 65^\circ = 360^\circ$ . ✓

**Answer:**  $\angle C = 102^\circ$ ,  $\angle D = 65^\circ$ .

## Common Mistakes

**X Confusing which arc subtends which angle — applying Theorem 2 to the wrong pair of angles (e.g.  $\angle AOB$  and  $\angle ACB$ ).**  
 OK Fix: Both angles must subtend the SAME arc (the same chord). Draw the chord connecting the two endpoints of the arc.



**X Applying the cyclic quadrilateral rule to ANY quadrilateral that happens to be inside a circle, not**

OK Fix: A cyclic quadrilateral must have all four vertices ON the circle. If a quadrilateral is inscribed with one vertex

**X Forgetting that the 'angle in a semicircle' theorem requires the chord to be a DIAMETER (passing**

OK Fix: If the problem says a triangle is inscribed in a semicircle, look for the diameter. The right angle is always

**X Thinking angles in the same segment are equal to the angle at the centre.**

OK Fix: Angles in the same segment are EQUAL to each other. They are HALF of the angle at the centre. Remember

## Summary

- The angle in a semicircle is always  $90^\circ$  (Thales' theorem).
- The angle at the centre of a circle is twice the angle at the circumference subtending the same arc.
- Angles in the same segment are equal to each other.
- A cyclic quadrilateral has all its vertices on the circle; opposite angles sum to  $180^\circ$ .
- The angle-in-a-semicircle theorem is a special case of the centre-twice-circumference theorem where the chord is a diameter.
- Always identify which arc a given angle subtends — this is the key to applying the theorems correctly.

## C1\_Q7: Circle Theorems 2: Tangents and Chords

### Hook

Imagine standing at the edge of a circular pond and looking straight ahead. The line of your gaze just touches the water at one point — that's a tangent. Now imagine throwing two ropes from the same point on the bank to two different points on the opposite shore — you'd be surprised to find both ropes are exactly the same length! These real-world intuitions are the foundation of some of the most tested circle theorems in IGCSE. Once you master them, you'll solve geometry problems that look impossible at first glance.

### Prerequisites

- C1\_Q6: Circle theorems 1 (angle in semicircle, centre-twice-circumference, angles in same segment, cyclic quadrilaterals)
- C1\_Q4: Triangle properties, isosceles triangles, congruence (RHS)
- Understanding of right angles and perpendicular lines
- Pythagoras' theorem (for chord length calculations)

### Core Explanation

#### ### Tangent-Radius Theorem

A **tangent** is a line that touches a circle at exactly one point (the **point of tangency**).

**\*\*The radius drawn to the point of tangency is perpendicular to the tangent.\*\***

If O is the centre, T is the point of tangency, and the line l is tangent to the circle at T, then:  
 $OT \perp l$

This means the angle between the radius and the tangent at the point of contact is  $90^\circ$ .

### Two Tangents from a Point

**\*\*From any point outside a circle, the two tangents drawn to the circle are equal in length.\*\***

If P is a point outside the circle, and tangents PA and PB touch the circle at A and B:  
 $PA = PB$

Furthermore, the line from P to the centre O bisects the angle between the two tangents:  
 $\angle APO = \angle BPO$

### Alternate Segment Theorem

**\*\*The angle between a tangent and a chord through the point of tangency is equal to the angle in the alternate segment.\*\***

If a tangent at A touches the circle, and AB is a chord, then for any point C on the circle on the opposite side of AB from the tangent:  
 $\angle \text{(tangent-chord at A)} = \angle ACB$

In other words, the angle between the tangent AX and chord AB equals the angle in the opposite (alternate) segment.

### Chord Properties

A **chord** is a straight line segment whose endpoints both lie on the circle.

**\*\*The perpendicular from the centre of a circle to a chord bisects the chord.\*\***

If O is the centre, AB is a chord, and  $OM \perp AB$ , then:  
 $AM = MB$

The distance from the centre to the chord is the perpendicular distance OM. If you know the radius r and the half-chord length AM, you can use Pythagoras:  
 $r^2 = OM^2 + AM^2$

**\*\*Conversely:\*\*** The line from the centre to the midpoint of a chord is perpendicular to the chord.

## Worked Example

**\*\*Problem 1:\*\*** A circle has centre O and radius 13 cm. A chord AB is 24 cm long. Find the perpendicular

distance from the centre to the chord.

**Step 1:** Draw the perpendicular from O to AB, meeting at M.

- By the chord theorem,  $OM \perp AB$  and  $AM = MB$ .

**Step 2:** Find AM.

$$AM = \frac{AB}{2} = \frac{24}{2} = 12 \text{ cm}$$

**Step 3:** Use Pythagoras' theorem in right triangle OMA.

-  $OA = 13$  cm (radius)

-  $AM = 12$  cm

$$OA^2 = OM^2 + AM^2$$

$$13^2 = OM^2 + 12^2$$

$$169 = OM^2 + 144$$

$$OM^2 = 25$$

$$OM = 5 \text{ cm}$$

**Answer:** The perpendicular distance from the centre to the chord is 5 cm.

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**Problem 2:** In the diagram, PA and PB are tangents from point P to a circle with centre O.  $\angle APB = 60^\circ$ . Find  $\angle AOB$ .

**Step 1:** Identify what we know.

- PA and PB are tangents, so  $OA \perp PA$  and  $OB \perp PB$  (tangent-radius theorem).

-  $\angle OAP = 90^\circ$  and  $\angle OBP = 90^\circ$ .

**Step 2:** Consider quadrilateral AOBP.

- Sum of interior angles in a quadrilateral =  $360^\circ$ .

$$\angle AOB + \angle OAP + \angle APB + \angle OBP = 360^\circ$$

$$\angle AOB + 90^\circ + 60^\circ + 90^\circ = 360^\circ$$

$$\angle AOB + 240^\circ = 360^\circ$$

$$\angle AOB = 120^\circ$$

**Answer:**  $\angle AOB = 120^\circ$ .

## Common Mistakes

**X Applying the alternate segment theorem backwards — using the angle between the tangent and**

OK Fix: The word 'alternate' means 'on the other side'. The angle between the tangent and chord equals the angle

**X Forgetting that the tangent-radius theorem creates a right angle — this is often the crucial step**

OK Fix: Whenever you see a tangent and radius at the same point, immediately mark the right angle ( $90^\circ$ )

**X Thinking the two tangents from a point are equal only for certain circles — it's true for ALL circles**

OK Fix: If you draw two tangents from any point outside any circle, the tangent segments are always equal in length

**X Confusing the chord bisector theorem — thinking the perpendicular from the centre bisects the c**

OK Fix: The perpendicular from the centre bisects the CHORD (the line segment), not the arc. The chord is split

## Summary

- The radius to a point of tangency is perpendicular to the tangent ( $OT \perp \text{tangent}$ ).
- Two tangents drawn from an external point to a circle are equal in length.
- The alternate segment theorem: the angle between a tangent and a chord equals the angle in the alternate segment.
- The perpendicular from the centre to a chord bisects the chord.
- Use Pythagoras' theorem with the radius, half-chord length, and perpendicular distance from the centre.
- These theorems often combine — for example, using tangent-radius right angles together with angle sums in quadrilaterals.